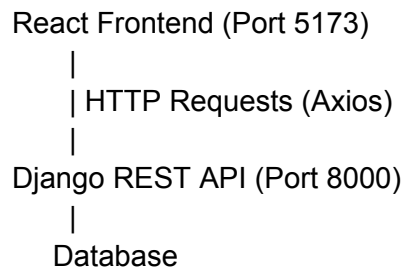


Architecture



Step 1: Start Django Server

Verify your API is working.

```
python manage.py runserver
```

Test:

```
http://127.0.0.1:8000/register/
```

```
http://127.0.0.1:8000/logout/
```

```
http://127.0.0.1:8000/books/
```

Step 2: Enable CORS in Django

Install:

```
pip install django-cors-headers
```

Add to `settings.py`

```
INSTALLED_APPS = [  
    ...  
    'corsheaders',  
]  
MIDDLEWARE = [  
    'corsheaders.middleware.CorsMiddleware',  
    ...  
]
```

Allow React URL:

```
CORS_ALLOWED_ORIGINS = [  
  "http://localhost:5173",  
]
```

Step 3: Create React Project

Using Vite:

```
npm create vite@latest frontend -- --template react
```

Move inside project:

```
cd frontend
```

Install dependencies:

```
npm install  
npm install axios react-router-dom
```

Start React:

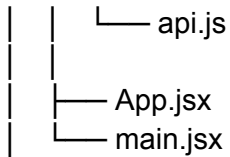
```
npm run dev
```

React runs on:

```
http://localhost:5173
```

Step 4: Project Structure

```
frontend/  
├── src/  
│   ├── pages/  
│   │   ├── Login.jsx  
│   │   ├── Register.jsx  
│   │   ├── Books.jsx  
│   │   └── EditBook.jsx  
│   └── services/
```



Step 5: Create Axios Configuration

`src/services/api.js`

```
import axios from "axios";

const API = axios.create({
  baseURL: "http://127.0.0.1:8000",
});

export default API;
```

Step 6: Create Register Page

`Register.jsx`

```
import { useState } from "react";
import API from "../services/api";

function Register() {
  const [formData, setFormData] = useState({
    username: "",
    password: "",
  });

  const handleSubmit = async (e) => {
    e.preventDefault();

    try {
      const response = await API.post("/register/", formData);
      alert("Registration Successful");
      console.log(response.data);
    } catch (error) {
      console.log(error.response.data);
    }
  };
}
```

```

    }
  };

  return (
    <form onSubmit={handleSubmit}>
      <input
        placeholder="Username"
        onChange={(e) =>
          setFormData({
            ...formData,
            username: e.target.value,
          })
        }
      />

      <input
        type="password"
        placeholder="Password"
        onChange={(e) =>
          setFormData({
            ...formData,
            password: e.target.value,
          })
        }
      />

      <button type="submit">Register</button>
    </form>
  );
}

export default Register;

```

Step 7: Create Books Page

Fetch all books.

`Books.jsx`

```

import { useEffect, useState } from "react";
import API from "../services/api";

```

```
function Books() {
  const [books, setBooks] = useState([]);

  const getBooks = async () => {
    const response = await API.get("/books/");
    setBooks(response.data);
  };

  useEffect(() => {
    getBooks();
  }, []);

  return (
    <>
      <h2>Books</h2>

      {books.map((book) => (
        <div key={book.id}>
          <h3>{book.title}</h3>
          <p>{book.author}</p>
        </div>
      ))}
    </>
  );
}

export default Books;
```

Step 8: Create Add Book

```
const addBook = async () => {
  await API.post("/books/", {
    title: "Python",
    author: "Sarath",
  });
};
```

Step 9: Delete Book

```
const deleteBook = async (id) => {
```

```
await API.delete(`/books/${id}/`);  
};
```

Step 10: Update Book

```
const updateBook = async (id) => {  
  await API.put(`/books/${id}/`, {  
    title: "Updated Title",  
    author: "Updated Author",  
  });  
};
```

Step 11: Setup React Router

Install:

```
npm install react-router-dom
```

`App.jsx`

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
```

```
import Register from "../pages/Register";
```

```
import Books from "../pages/Books";
```

```
function App() {  
  return (  
    <BrowserRouter>  
      <Routes>  
        <Route path="/" element={<Books />} />  
        <Route path="/register" element={<Register />} />  
      </Routes>  
    </BrowserRouter>  
  );  
}
```

```
export default App;
```